

Programme	B. Sc. STATISTICS				
Course Code	STA 8 CJ 408				
Course Title	GENERALIZED LINEAR MODELS				
Type of Course	Major				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Elementary ideas about linear estimation.				
Course Summary	Understand about generalized linear models.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
1.	Identify the general theory of GLM	U	C	Seminar Presentation /
2.	Describe the use of binary and multinomial Understand and logistic models and apply them for various data sets	R	F	Instructor-created exams
3.	Explain the concepts related to count data GLM and apply them for various count data sets.	Ap	C	Group Tutorial Work
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (60)	Marks (70)
I	Components of a generalized linear model (GLM)		10	15
	1	Random component		
	2	linear predictor, link function		
	3	Quantitative/qualitative explanatory variables and interpreting effects		
	4	Model matrices and model vector spaces		
	5	Identifiability and estimability		
II	Generalized linear models		12	20
	6	Model fitting and inference		

	7	Exponential dispersion family distributions		
	8	Likelihood and asymptotic distributions		
	9	Likelihood-ratio/Wald/Score methods of inference		
	10	Parameters, deviance, model comparison, and model checking		
	11	Goodness of fit		
III	Binary logistic models, nominal responses		10	15
	12	Baseline-category logit models		
	13	Ordinal responses: cumulative logit and probit models		
	14	Probit and complementary log–log models,		
	15	Multinomial response models		
IV	Models for count data		16	20
	16	Poisson GLMs for counts and rates		
	17	Poisson/multinomial models for contingency tables		
	18	Negative Binomial GLMS		
	19	Models for zero-inflated data		
	20	Quasi-likelihood methods		
	21	Variance inflation for over dispersed Poisson and Binomial GLMs		
	22	Beta-Binomial models and Quasi-likelihood alternatives		
	23	Quasi-likelihood and model misspecification		
V	Open –Ended		12	
	Model building and validation in practical situations using R software			

Reference

1. **Agresti, A. (2015).** Foundations of Linear and Generalized Linear Models, Wiley
2. **Dobson, A. J. (2002).** An Introduction to Generalized Linear Models, 2nd Ed. Chapman & Hall
3. **Jiang, J. (2007).** Linear and Generalized Linear Mixed Models and their Applications, Springer
4. **Jong, P. and Heller, G. Z. (2008)** Generalized Linear Models for Insurance Data, Cambridge University Press.
5. **Lindsey, J. K. (1997).** Applying Generalized Linear Models, Springer
6. **McCullagh, P. and Nelder, J. A. (1989).** Generalized Linear Models, Chapman & Hall
7. **McCulloch, C. E. and Searle, S. R. (2001).** Generalized, Linear and Mixed Models, Wiley
8. **Stroup, W. W. (2013).** Generalized Linear Mixed Models, Modern Concepts, Methods and Applications, CRC Press